DATABASE MANAGEMENT SYSTEM LAB

ASSIGNMENT NO 2

DDL AND DML QUERIES

Case Study :- Airline Management System

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DDL COMMANDS

```
mysql> CREATE DATABASE airline_management;
Query OK, 1 row affected (0.04 sec)

mysql> USE airline_management;
Database changed
mysql> CREATE TABLE Airline(
    -> airline_id INT,
    -> airline_name VARCHAR(50),
    -> country VARCHAR(40)
    -> );
Query OK, 0 rows affected (0.10 sec)

mysql> CREATE TABLE Flight(
    -> flight_id INT,
    -> airline_id INT,
    -> source VARCHAR(40),
    -> destination VARCHAR(40),
    -> departure_time TIME,
    -> arrival_time TIME,
    -> flight_type VARCHAR(15),
    -> total_seats INT,
    -> status VARCHAR(15)
    -> );
Query OK, 0 rows affected (0.05 sec)
```

```
mysql> CREATE TABLE Passenger(
    -> passenger_id INT,
    -> passenger_name VARCHAR(50),
    -> gender CHAR(1),
    -> age INT,
    -> passport_no VARCHAR(20),
    -> nationality VARCHAR(40)
    -> );
Query OK, 0 rows affected (0.06 sec)

mysql> CREATE TABLE Booking (
    ->     booking_id INT,
    ->     flight_id INT,
    ->     passenger_id INT,
    ->     booking_date DATE,
    ->     seat_no VARCHAR(5),
    ->     fare DECIMAL(10,2),
    ->     travel_class VARCHAR(15)
    -> );
Query OK, 0 rows affected (0.06 sec)
```

```
mysql> DESC Airline;
+--------------+-------------+------+-----+---------+-------+
| Field        | Type        | Null | Key | Default | Extra |
+--------------+-------------+------+-----+---------+-------+
| airline_id   | int         | YES  |     | NULL    |       |
| airline_name | varchar(50) | YES  |     | NULL    |       |
| country      | varchar(40) | YES  |     | NULL    |       |
+--------------+-------------+------+-----+---------+-------+
3 rows in set (0.05 sec)

mysql> DESC Flight;
+----------------+-------------+------+-----+---------+-------+
| Field          | Type        | Null | Key | Default | Extra |
+----------------+-------------+------+-----+---------+-------+
| flight_id      | int         | YES  |     | NULL    |       |
| airline_id     | int         | YES  |     | NULL    |       |
| source         | varchar(40) | YES  |     | NULL    |       |
| destination    | varchar(40) | YES  |     | NULL    |       |
| departure_time | time        | YES  |     | NULL    |       |
| arrival_time   | time        | YES  |     | NULL    |       |
| flight_type    | varchar(15) | YES  |     | NULL    |       |
| total_seats    | int         | YES  |     | NULL    |       |
| status         | varchar(15) | YES  |     | NULL    |       |
+----------------+-------------+------+-----+---------+-------+
9 rows in set (0.00 sec)

mysql> DESC Passenger;
+----------------+-------------+------+-----+---------+-------+
| Field          | Type        | Null | Key | Default | Extra |
+----------------+-------------+------+-----+---------+-------+
| passenger_id   | int         | YES  |     | NULL    |       |
| passenger_name | varchar(50) | YES  |     | NULL    |       |
| gender         | char(1)     | YES  |     | NULL    |       |
| age            | int         | YES  |     | NULL    |       |
| passport_no    | varchar(20) | YES  |     | NULL    |       |
| nationality    | varchar(40) | YES  |     | NULL    |       |
+----------------+-------------+------+-----+---------+-------+
6 rows in set (0.02 sec)

mysql> DESC Booking;
+--------------+---------------+------+-----+---------+-------+
| Field        | Type          | Null | Key | Default | Extra |
+--------------+---------------+------+-----+---------+-------+
| booking_id   | int           | YES  |     | NULL    |       |
| flight_id    | int           | YES  |     | NULL    |       |
| passenger_id | int           | YES  |     | NULL    |       |
| booking_date | date          | YES  |     | NULL    |       |
| seat_no      | varchar(5)    | YES  |     | NULL    |       |
| fare         | decimal(10,2) | YES  |     | NULL    |       |
| travel_class | varchar(15)   | YES  |     | NULL    |       |
+--------------+---------------+------+-----+---------+-------+
7 rows in set (0.00 sec)
```

```
mysql> ALTER TABLE Airline
    -> ADD established_year YEAR;
Query OK, 0 rows affected (0.06 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> DESC Airline;
+------------------+-------------+------+-----+---------+-------+
| Field            | Type        | Null | Key | Default | Extra |
+------------------+-------------+------+-----+---------+-------+
| airline_id       | int         | YES  |     | NULL    |       |
| airline_name     | varchar(50) | YES  |     | NULL    |       |
| country          | varchar(40) | YES  |     | NULL    |       |
| established_year | year        | YES  |     | NULL    |       |
+------------------+-------------+------+-----+---------+-------+
4 rows in set (0.00 sec)

mysql> ALTER TABLE Booking
    -> DROP fare;
Query OK, 0 rows affected (0.05 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> DESC Booking;
+--------------+-------------+------+-----+---------+-------+
| Field        | Type        | Null | Key | Default | Extra |
+--------------+-------------+------+-----+---------+-------+
| booking_id   | int         | YES  |     | NULL    |       |
| flight_id    | int         | YES  |     | NULL    |       |
| passenger_id | int         | YES  |     | NULL    |       |
| booking_date | date        | YES  |     | NULL    |       |
| seat_no      | varchar(5)  | YES  |     | NULL    |       |
| travel_class | varchar(15) | YES  |     | NULL    |       |
+--------------+-------------+------+-----+---------+-------+
6 rows in set (0.01 sec)
```

DML COMMANDS

```
mysql> INSERT INTO Airline VALUES
    -> (1, 'Air India', 'India'),(2,'Indigo','India'),(3,'Spicejet','India'),(4,'Airindia','India');
Query OK, 4 rows affected (0.02 sec)
Records: 4  Duplicates: 0  Warnings: 0
       INSERT INTO Flight VALUES
    -> (101, 1, 'Mumbai', 'Delhi', '10:00', '12:00', 'Domestic', 180, 'Scheduled'),
    -> (102,2,'kolkata','nagpur','12:00','4:00','Domestic',180,'Scheduled'),
    -> (103,3,'pune','banglore','3:00','8:00','Domestic',180,'Scheduled'),
    -> (104,4,'mumbai','chennai','6:00','9:00','Domestic',180,'Scheduled');
Query OK, 4 rows affected (0.02 sec)
Records: 4  Duplicates: 0  Warnings: 0

mysql> INSERT INTO Passenger VALUES
    -> (501, 'Rahul Mehta', 'M', 35, 'P1234567', 'Indian'),
    -> (301,'niharika','F',34,'P788999','Indian'),
    -> (678,'Rohit','M',23,'P85666','Indian'),
    -> (888,'Jiya','F',46,'P45677','Indian');
Query OK, 4 rows affected (0.03 sec)
Records: 4  Duplicates: 0  Warnings: 0

mysql> INSERT INTO Booking VALUES
    -> (1001, 101, 501, '2024-02-01', '12A', 5500, 'Economy'),
    -> (1002,105,789,'2023-09-08','14B',9000,'Economy'),
    -> (1003,123,890,'2025-07-07','13A',7888,'Economy');
Query OK, 3 rows affected (0.03 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

```
mysql> SELECT * FROM Ailrine;
ERROR 1146 (42S02): Table 'airline_management.ailrine' doesn't exist
mysql> select * from Airline;
+------------+--------------+---------+--------------+
| airline_id | airline_name | country | founded_year |
+------------+--------------+---------+--------------+
|          1 | Air India    | India   |         NULL |
|          2 | Indigo       | India   |         NULL |
|          3 | Spicejet     | India   |         NULL |
|          4 | Airindia     | India   |         NULL |
+------------+--------------+---------+--------------+
4 rows in set (0.00 sec)

mysql> select * from flight;
+-----------+------------+---------+-------------+----------------+--------------+-------------+-------------+-----------+
| flight_id | airline_id | source  | destination | departure_time | arrival_time | flight_type | total_seats | status    |
+-----------+------------+---------+-------------+----------------+--------------+-------------+-------------+-----------+
|       101 |          1 | Mumbai  | Delhi       | 10:00:00       | 12:00:00     | Domestic    |         180 | Scheduled |
|       102 |          2 | kolkata | nagpur      | 12:00:00       | 04:00:00     | Domestic    |         180 | Scheduled |
|       103 |          3 | pune    | banglore    | 03:00:00       | 08:00:00     | Domestic    |         180 | Scheduled |
|       104 |          4 | mumbai  | chennai     | 06:00:00       | 09:00:00     | Domestic    |         180 | Scheduled |
+-----------+------------+---------+-------------+----------------+--------------+-------------+-------------+-----------+
4 rows in set (0.00 sec)

mysql> SELECT flight_id,source  FROM Flight;
+-----------+---------+
| flight_id | source  |
+-----------+---------+
|       101 | Mumbai  |
|       102 | kolkata |
|       103 | pune    |
|       104 | mumbai  |
+-----------+---------+
4 rows in set (0.00 sec)

mysql> SELECT * FROM Flight
    -> WHERE flight_id>102;
+-----------+------------+--------+-------------+----------------+--------------+-------------+-------------+-----------+
| flight_id | airline_id | source | destination | departure_time | arrival_time | flight_type | total_seats | status    |
+-----------+------------+--------+-------------+----------------+--------------+-------------+-------------+-----------+
|       103 |          3 | pune   | banglore    | 03:00:00       | 08:00:00     | Domestic    |         180 | Scheduled |
|       104 |          4 | mumbai | chennai     | 06:00:00       | 09:00:00     | Domestic    |         180 | Scheduled |
+-----------+------------+--------+-------------+----------------+--------------+-------------+-------------+-----------+
2 rows in set (0.02 sec)
```

```
mysql> SELECT * FROM Flight
    -> WHERE flight_id>103 AND flight_type='Domestic';
+-----------+------------+--------+-------------+----------------+--------------+-------------+-------------+-----------+
| flight_id | airline_id | source | destination | departure_time | arrival_time | flight_type | total_seats | status    |
+-----------+------------+--------+-------------+----------------+--------------+-------------+-------------+-----------+
|       104 |          4 | mumbai | chennai     | 06:00:00       | 09:00:00     | Domestic    |         180 | Scheduled |
+-----------+------------+--------+-------------+----------------+--------------+-------------+-------------+-----------+
1 row in set (0.02 sec)

mysql> SELECT * FROM Airline
    -> WHERE airline_name LIKE 'A%';
+------------+--------------+---------+--------------+
| airline_id | airline_name | country | founded_year |
+------------+--------------+---------+--------------+
|          1 | Air India    | India   |         NULL |
|          4 | Airindia     | India   |         NULL |
+------------+--------------+---------+--------------+
2 rows in set (0.02 sec)

mysql> SELECT * FROM Airline
    -> WHERE airline_id BETWEEN 1 AND 4;
+------------+--------------+---------+--------------+
| airline_id | airline_name | country | founded_year |
+------------+--------------+---------+--------------+
|          1 | Air India    | India   |         NULL |
|          2 | Indigo       | India   |         NULL |
|          3 | Spicejet     | India   |         NULL |
|          4 | Airindia     | India   |         NULL |
+------------+--------------+---------+--------------+
4 rows in set (0.01 sec)

mysql> SELECT * FROM Passenger
    -> ;
ERROR 1146 (42S02): Table 'airline_management.passenger' doesn't exist
mysql> SELECT * FROM Flight
    -> ORDER BY flight_id DESC;
+-----------+------------+---------+-------------+----------------+--------------+-------------+-------------+-----------+
| flight_id | airline_id | source  | destination | departure_time | arrival_time | flight_type | total_seats | status    |
+-----------+------------+---------+-------------+----------------+--------------+-------------+-------------+-----------+
|       104 |          4 | mumbai  | chennai     | 06:00:00       | 09:00:00     | Domestic    |         180 | Scheduled |
|       103 |          3 | pune    | banglore    | 03:00:00       | 08:00:00     | Domestic    |         180 | Scheduled |
|       102 |          2 | kolkata | nagpur      | 12:00:00       | 04:00:00     | Domestic    |         180 | Scheduled |
|       101 |          1 | Mumbai  | Delhi       | 10:00:00       | 12:00:00     | Domestic    |         180 | Scheduled |
+-----------+------------+---------+-------------+----------------+--------------+-------------+-------------+-----------+
4 rows in set (0.00 sec)
```

```
mysql>  UPDATE Flight
    -> SET destination='lucknow'
    -> WHERE flight_id=101;
Query OK, 1 row affected (0.02 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> DELETE FROM Flight
    -> WHERE flight_id=103;
Query OK, 1 row affected (0.02 sec)
```